

Pressure in Solids and Liquids — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Award method marks for a correct formula and a correct substitution even when the arithmetic fails. Almost every mark lost on this topic goes one of three ways: an area left in square centimetres and the answer still called pascals, the belief that a larger force must always mean a larger pressure, or the belief that pressure in a liquid depends on how much liquid there is. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 pressure = force ÷ area **[1]** [2]
Unit: the **pascal (Pa)**, which is the same as N/m² **[1]**
N/m without the square is not the unit of pressure and scores nothing here.
- 2 (a) $200 \div 4 = \mathbf{50\ Pa}$ [4]
(b) $60 \div 0.2 = \mathbf{300\ Pa}$
(c) $750 \div 0.05$ **[1] = 15 000 Pa [1]**
Multiplying instead of dividing gives 800 in (a). Check the direction of the formula before the arithmetic.
- 3 (a) $F = p \times A = 500 \times 6 = \mathbf{3000\ N}$ [2]
(b) $20\ 000 \times 0.02 = \mathbf{400\ N}$
- 4 (a) $A = F \div p = 600 \div 300 = \mathbf{2\ m^2}$ [2]
(b) $1500 \div 30\ 000 = \mathbf{0.05\ m^2}$
The unit here is m², not Pa. Carrying the wrong unit across from the previous question is common.
- 5 (a) $20 \div 10\ 000 = \mathbf{0.002\ m^2}$ [3]
(b) $400 \div 10\ 000 = \mathbf{0.04\ m^2}$
(c) $2.5 \div 10\ 000 = \mathbf{0.00025\ m^2}$
Dividing by 100 instead of 10 000 is the error to look for. It gives answers a hundred times too big every time.
- 6 $20\ \text{cm}^2 = 0.002\ \text{m}^2$ **[1]** [3]
 $60 \div 0.002$ **[1]**
 $= \mathbf{30\ 000\ Pa [1]}$
 $60 \div 20 = 3$ scores nothing beyond the substitution mark. That answer is newtons per square centimetre, and it is the single most common error in this topic.
- 7 Area = $0.4 \times 0.5 = 0.2\ \text{m}^2$ **[1]** [3]
 $480 \div 0.2$ **[1]**
 $= \mathbf{2400\ Pa [1]}$
The area line is worth a mark on its own. A candidate who writes only the final number risks two of the three marks.
- 8 Wide tyres give a larger contact area with the ground **[1]** [2]
Pressure = force ÷ area, so for the same weight a larger area gives a smaller pressure and the tractor does not sink into soft soil **[1]**
“So it does not sink” alone is one mark. The second needs area and pressure named and linked.

- 9** The point has a very small area, so the same force gives a very large pressure and the pin goes into the board **[1] [1]** [3]
The disc has a large area, so the pressure on the thumb is small and it does not hurt **[1]**
One force, two areas, two very different pressures. A candidate who says the point is “sharper so stronger” has not used the physics.
- 10** $p = \rho \times g \times h$ **[1]** [3]
 $= 1000 \times 10 \times 8$ **[1]**
 $= 80\,000\text{ Pa}$ (80 kPa) **[1]**
- 11** Depends on: the **depth** below the surface **[1]** [3]
Depends on: the **density** of the liquid (accept g) **[1]**
Does not depend on: the volume, the width, the shape of the container or the total amount of liquid **[1]**
This question is the written form of the trap tested in Papers B and D. Anything naming volume or shape as a factor loses the third mark.

Paper B — Medium

30 MARKS

- 1** (a) $600 \div 0.03$ **[1] = 20 000 Pa [1]** [5]
(b) One foot is 0.015 m^2 , so $600 \div 0.015$ **[1] = 40 000 Pa [1]**
(c) The force is **unchanged** at 600 N — it is still his whole weight **[1]**
The commonest error in (b) is halving the force as well as the area, giving 20 000 Pa again. Lifting a foot does not make him lighter.
- 2** (a) $4 \times 25 = 100\text{ cm}^2$ **[1]**; $100 \div 10\,000 = 0.01\text{ m}^2$ **[1]** [5]
(b) $900 \div 0.01$ **[1] = 90 000 Pa [1]**
(c) $900 \div 0.3 = 3000\text{ Pa}$ **[1]**
Using one foot instead of four gives 360 000 Pa. The board reduces the pressure by a factor of 30 while removing no force at all — worth pointing out when marking.
- 3** (a) $1000 \times 10 \times 3$ **[1] = 30 000 Pa [1]** [4]
(b) $1030 \times 10 \times 20$ **[1] = 206 000 Pa [1]**
Rounding 1030 to 1000 gives 200 000 and loses the accuracy mark. The question supplied the exact density deliberately.
- 4** Pressure in a liquid depends only on depth, density and g **[1]** [4]
 $p = \rho gh$ contains no volume, width or shape, so the amount of water is irrelevant **[1]**
 $1000 \times 10 \times 0.5$ **[1]**
 $= 5000\text{ Pa}$ in **both** containers **[1]**
The fourth mark requires the word “both”, or two identical answers. Calculating one pressure and stopping scores three.
- 5** Pressure in a liquid increases with depth **[1]** [3]
so the water pushes hardest at the base of the dam and hardly at all at the top **[1]**
The wall is made thicker there so it is strong enough to withstand the greater pressure **[1]**
Describing the shape is not explaining it. Without the words “increases with depth” this caps at one mark.
- 6** Water squirts out **sideways** from all three holes **[1]** [4]
because pressure in a liquid acts equally in all directions, not only downwards **[1]**
The **lowest** jet travels furthest **[1]**
because the pressure is greater at greater depth **[1]**
Two separate ideas, two marks each. Candidates usually get the depth half and miss the all-directions half entirely.

- 7 Atmospheric pressure is the pressure exerted by the weight of the air above a surface [3]
[1]
It is about 100 000 Pa (100 kPa) at sea level [1]
It is not noticed because it acts equally in all directions and the pressure inside the body pushes back equally hard [1]
- 8 Water pressure = $1000 \times 10 \times 12 = 120\,000\text{ Pa}$ [1] [2]
Total = $120\,000 + 100\,000 = 220\,000\text{ Pa}$ [1]
Omitting the atmosphere is the error the question is built to catch. The word “total” is the signal to add it.

Paper C — Hard

30 MARKS

- 1 (a) Area = $1.5 \times 0.8 = 1.2\text{ m}^2$; $2400 \div 1.2$ [1] = **2000 Pa** [1] [5]
(b) Area = $0.8 \times 0.5 = 0.4\text{ m}^2$; $2400 \div 0.4$ [1] = **6000 Pa** [1]
(c) The pressure is **three times** greater, because the area is three times smaller while the weight is unchanged [1]
The block did not get heavier. Any answer in (c) that refers to a change in force is wrong even if the factor of 3 is stated.
- 2 (a) $0.15 \div 10\,000$ [1] = **0.000015 m²** [1] [5]
(b) $45 \div 0.000015$ [1] = **3 000 000 Pa** (3 MPa) [1]
(c) Ten times the area, so a tenth of the pressure: **300 000 Pa** [1]
A sharp knife is not a stronger knife. If a candidate says the blunt knife needs more force, they have inverted the whole idea — award (c) only for 300 000 Pa.
- 3 (a) $1030 \times 10 \times 60$ [1] = **618 000 Pa** [1] [6]
(b) $618\,000 + 100\,000$ [1] = **718 000 Pa** [1]
(c) $F = p \times A = 718\,000 \times 0.05$ [1] = **35 900 N** [1]
Using 618 000 in (c) gives 30 900 N and scores the method mark only. Follow-through applies if (b) itself was wrong.
- 4 (a) X: $1000 \times 10 \times 4 = 40\,000\text{ Pa}$ [1]; Y: $1000 \times 10 \times 12 = 120\,000\text{ Pa}$ [1] [5]
(b) $120\,000 - 40\,000 = 80\,000\text{ Pa}$ [1]
(c) **Nothing changes** [1] — pressure depends on depth and density only, and neither has changed; the surface area of the water does not appear in $p = \rho gh$ [1]
Part (c) is the discriminator on this paper. Three times the depth gives three times the pressure, and a tenth of the area gives exactly the same pressure.
- 5 (a) $20 \div 0.0004$ [1] = **50 000 Pa** [1] [5]
(b) The liquid transmits the same pressure to the large piston; $F = 50\,000 \times 0.03$ [1] = **1500 N** [1]
(c) **$\times 75$** , which is the ratio of the two areas [1]
The idea being tested is that the pressure is the same at both pistons. A candidate who recalculates a different pressure for the large piston has missed the point of hydraulics.
- 6 A liquid is (almost) incompressible [1] [4]
so pressure applied at one piston is transmitted undiminished to the other [1]
A gas would be compressed by the applied force [1]
so the small piston would move without the large piston moving, and the force transmitted would be reduced [1]
“Gases are squashy” earns the third mark only. The fourth needs the consequence for the machine.

Paper D — Hard

30 MARKS

- 1** (a) Total area = $4 \times 0.1 = 0.4 \text{ m}^2$ **[1]**; $40\,000 \div 0.4 = 100\,000 \text{ Pa}$ **[1]** [6]
 (b) $500 \div 0.0001$ **[1] = 5 000 000 Pa [1]**
 (c) Pressure is force divided by area, not force alone **[1]**; she is 80 times lighter (500 N against 40 000 N), but her heel spreads that weight over 0.0001 m^2 against the elephant's **total** 0.4 m^2 — an area 4000 times smaller. $4000 \div 80 = 50$, which is exactly the ratio of the two pressures **[1]**
Using one foot rather than four in (a) gives 400 000 Pa. This question exists to break the belief that a bigger force always means a bigger pressure.
- 2** (a) $A = F \div p$ **[1] = $1800 \div 4000$ [1] = 0.45 m^2 [1]** [5]
 (b) $0.45 \div 2 = 0.225 \text{ m}^2$ each **[1]**
 (c) $2250 \div 0.45 = 5000 \text{ Pa}$, which is above the 4000 Pa limit, so it sinks **[1]**
Multiplying instead of dividing in (a) gives $7\,200\,000 \text{ m}^2$, an area larger than a football pitch. Any candidate who writes that without noticing has stopped checking whether answers are sensible.
- 3** (a) $1000 \times 10 \times 2.5$ **[1] = $25\,000 \text{ Pa}$ [1]** [5]
 (b) Still **$25\,000 \text{ Pa}$ [1]**; the depth and the density are unchanged, and $p = \rho gh$ contains no width, shape or volume term **[1]**
 (c) Equally in **all directions** — downwards, sideways and upwards **[1]**
Halving the width halves the amount of water and changes nothing at all. If a candidate halves the pressure in (b), they are using volume as a factor.
- 4** (a) $13\,600 \times 10 \times 0.75$ **[1] = $102\,000 \text{ Pa}$ [1]** [4]
 (b) That is almost exactly atmospheric pressure **[1]**, so the air pressure will support a mercury column about this tall, and the height of the column can be read directly as a measure of the pressure **[1]**
Mercury is used because it is dense: the same pressure would need about 10 m of water, which is why barometers are not made of water.
- 5** (a) Water: $1000 \times 10 \times 5 = 50\,000 \text{ Pa}$ **[1]**; oil: $800 \times 10 \times 5 = 40\,000 \text{ Pa}$ **[1]** [4]
 (b) $h = p \div (\rho g) = 50\,000 \div (800 \times 10)$ **[1] = 6.25 m [1]**
The oil is 80% as dense, so it needs $1/0.8 = 1.25$ times the depth. A candidate who answers 4 m has used the ratio upside down.
- 6** (i) The mistake: dividing by an area in cm^2 and calling the answer pascals **[1]** [6]
 $20 \text{ cm}^2 = 20 \div 10\,000 = 0.002 \text{ m}^2$ **[1]**
 $p = 60 \div 0.002 = 30\,000 \text{ Pa}$ **[1]**
 (ii) The mistake: pressure in a liquid does not depend on the amount of liquid **[1]**
 $p = \rho gh$ depends only on depth, density and g , and both tanks are 2 m deep **[1]**
 Both are **$20\,000 \text{ Pa}$ [1]**
These are the two errors this whole topic is built around. A candidate who can diagnose both here will not make them under exam pressure — that is the purpose of the question.